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Hot-stamped product

Abstract

A hot-stamped product with improved delayed-fracture resistance is provided. The hot-stamped product **1** includes: a steel substrate **10**; and an Al film **20** formed on the steel substrate **10**, the Al film **20** including: an interface layer **21** located at the interface with the steel substrate **10** and with part of the α Fe substituted by Al and Si; an intermediate layer **22** formed on the interface layer **21**; and an oxide layer formed on the intermediate layer, the intermediate layer **22** including an Fe—Al—Si phase **22a** with part of the α Fe substituted by Al and Si, the Fe—Al—Si phase **22a** including one or more elements selected from the group consisting of Zr, Ce, Y, Ta, Ni, Cu, Nb, Cr, Co, V and Ti, the oxide layer **23** including one or more elements selected from the group consisting of Be, Mg, Ca, Sr, Ba, Sc and Zn.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a hot-stamped product.

BACKGROUND ART

(2) One known method of forming a high-strength steel member with high size precision is hot stamping (also referred to as hot pressing, die quenching, press quenching, etc.). Hot stamping involves heating steel material, such as steel sheet, to the austenite region and performing hot forming before cooling the formed material to provide desired properties.

(3) In hot stamping, scale may be produced on the surface of the steel material during heating, and that must be removed by a post-process. A technique to avoid this is known, where steel material to be hot stamped is Al-plated to prevent formation of scale.

(4) JP S63-3929 B2 discloses a method of manufacturing a molten-aluminum-plated steel sheet exhibiting reduced values of the oxidation mass gain during high-temperature oxidation. Japanese Patent No. 2943021 discloses a method of manufacturing an austenitic stainless steel sheet strip having an NiAl intermetallic compound at its surface layer, where the surface of an austenitic stainless steel sheet strip is coated with Al and then diffusion heat treatment is performed in a non-oxidizing atmosphere within a temperature range of 700° C. to 800° C.

(5) WO 2018/221738 A1 discloses a hot-stamped member having good post-coating corrosion resistance. This hot-stamped member includes steel material, an Al—Fe intermetallic compound layer, and an oxide layer, the oxide layer containing one or more elements selected from the group consisting of Be, Mg, Ca, Sr, Ba, Sc, Ti, V, Cr, Mn, Fe, Co, Ni, Cu and Zn in a ratio, excluding oxygen, not lower than 0.01 at. % and not higher than 80 at. %.

(6) JP 2017-536472 A discloses a flat steel product to be hot formed including a steel substrate and a protection film mainly composed of Al. This publication discloses that the protection film contains at least one alkali earth metal or transition metal in a total of 0.1 weight % to 0.5 weight %, where oxides of the alkali earth metal(s) or transition metal(s) are formed on the surface of the protection film during hot forming.

PRIOR ART DOCUMENTS

Patent Documents

(7) Patent Document 1: JP S63-3929 B2 Patent Document 2: Japanese Patent No. 2943021 Patent Document 3: WO 2018/221738 A1 Patent Document 4: JP 2017-536472 A

SUMMARY OF THE INVENTION

Problems to be Solved by the Invention

(8) When an Al-plated steel material is hot stamped, oxygen and water in the atmosphere are consumed during oxidation of Al. Hydrogen emitted during this process may be absorbed in the steel material, potentially causing delayed fracture.

(9) A problem to be solved by the present invention is to provide a hot-stamped product with improved delayed-fracture resistance.

Means for Solving the Problems

(10) A hot-stamped product according to an embodiment of the present invention includes: a steel substrate; and an Al film formed on the steel substrate, the Al film including: an interface layer located at an interface with the steel substrate and having a structure with part of α Fe substituted by Al and Si; an intermediate layer formed on the interface layer; and an oxide layer formed on the intermediate layer, a thickness of the intermediate layer being not smaller than 15 μ m, the intermediate layer including an Fe—Al—Si phase having a structure with part of α Fe substituted

by Al and Si, the Fe—Al—Si phase including one or more elements selected from the group consisting of Zr, Ce, Y, Ta, Ni, Cu, Nb, Cr, Co, V and Ti, an Si content of the Fe—Al—Si phase being 1 to 20 mass %, a total content of Zr, Ce, Y, Ta, Ni, Cu, Nb, Cr, Co, V and Ti being 0.10 to 5.0 mass %, the oxide layer including one or more elements selected from the group consisting of Be, Mg, Ca, Sr, Ba, Sc and Zn, a total proportion of Be, Mg, Ca, Sr, Ba, Sc and Zn in a composition of the oxide layer excluding oxygen being 0.01 to 80.0 mass %.

Effects of the Invention

(11) The present invention provides a hot-stamped product with improved delayed-fracture resistance.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a cross-sectional view of a plated steel member prior to hot stamping, schematically illustrating its structure.

(2) FIG. 2 is a cross-sectional view of an exemplary hot-stamped product formed by hot stamping the plated steel member, schematically illustrating its structure.

(3) FIG. 3 is a cross-sectional view of a hot-stamped product according to an embodiment of the present invention, schematically illustrating its structure.

(4) FIG. 4 is a photograph showing a cross section of the Al film of a product labeled a1.

(5) FIG. 5 is a photograph of a cross section of the Al film of a product labeled A13.

EMBODIMENTS FOR CARRYING OUT THE INVENTION

(6) The present inventors conducted a variety of investigations on hydrogen absorption during hot stamping of an Al-plated steel member, particularly an Al—Si plated steel member.

(7) FIG. 1 is a cross-sectional view of a plated steel member **1A** prior to hot stamping, schematically illustrating its structure. The plated steel member **1A** includes a steel substrate **10** and an Al—Si plating layer **30**. The plated steel member **1A** further includes an Al—Si—Fe alloy layer **35** between the steel substrate **10** and Al—Si plating layer **30**, formed by Fe in the steel substrate **10** diffusing into the plating layer.

(8) FIG. 2 is a cross-sectional view of a hot-stamped product **9**, which is an exemplary hot-stamped product formed by hot stamping the plated steel member **1A**, schematically illustrating its structure. The hot-stamped product **9** includes the steel substrate **10** and an Al film **20** formed on the steel substrate **10**.

(9) The Al film **20** includes a plurality of layers. Specifically, the Al film **20** includes an interface layer **21** located at the interface with the steel substrate **10**, an intermediate layer **22** formed on the interface layer **21**, and an oxide layer **23** formed on the intermediate layer **22**. It is noted that the intermediate layer **22** includes an Fe—Al—Si phase **22a** and an Al—Fe phase **22b** with different structures, where the Fe—Al—Si phase **22a** and Al—Fe phase **22b** may be composed of (sub-)layers. Further, the above-stated classification is based on appearance during observation with an optical microscope or a scanning electron microscope (SEM); if observed with a transmission electron microscope (TEM), each layer may be observed as a collection of several crystal phases.

(10) The interface layer **21** is derived from the Al—Si—Fe alloy layer **35** of the plated steel member **1A** (FIG. 1), and is mainly structured such that part of the bcc structure of the α Fe is substituted by Al and Si. The interface layer **21** may contain an intermediate phase, such as τ phase.

(11) The intermediate layer **22** includes the Fe—Al—Si phase **22a** and Al—Fe phase **22b**. As is the case with the interface layer **21**, the Fe—Al—Si phase **22a** is structured such that part of the bcc structure of the α Fe is substituted by Al and Si. The Al—Fe phase **22b** has a structure of Fe.sub.4Al.sub.13 and Fe.sub.2Al.sub.5. As shown in FIG. 2, the Fe—Al—Si phase **22a** typically

has the shape of a layer (i.e., band), located in the middle of the intermediate layer **22** as determined along the thickness direction and sandwiched between the upper and lower layers of the Al—Fe phase **22b**.

(12) The oxide layer **23** is mainly composed of oxides of Al.

(13) The present inventors discovered that hydrogen absorption by the steel substrate **10** is reduced if the Fe—Al—Si phase **22a** contains predetermined amounts of specific elements, specifically one or more of Zr, Ce, Y, Ta, Ni, Cu, Nb, Cr, Co, V and Ti (these eleven elements will be hereinafter referred to as “Group A elements”).

(14) FIG. **3** is a cross-sectional view of a hot-stamped product **1**, which is an exemplary hot-stamped product where the Fe—Al—Si phase **22a** contains predetermined amounts of Group A elements, schematically illustrating its structure. In the hot-stamped product **1**, the size of the Fe—Al—Si phase **22a** is smaller than in the hot-stamped product **9** (FIG. **2**). In the hot-stamped product **9** (FIG. **2**), the Fe—Al—Si phase **22a** has the shape of a layer (i.e., band), whereas the Fe—Al—Si phase **22a** of the hot-stamped product **1** is discontinuous. Further, the Group A elements are selectively distributed in the Fe—Al—Si phase **22a**.

(15) This suggests that the Group A elements were selectively distributed in the Fe—Al—Si phase **22a** during heating, and also have the effect of reducing stabilization of the Fe—Al—Si phase **22a**.

(16) As the Fe—Al—Si phase **22a** is prevented from stabilizing, growth of the Al—Fe phase **22b** is promoted. As the thickness for the Al—Fe phase **22b** increases, Al in the Al film **20** is prevented from diffusing toward its surface, thus reducing oxidation of Al. This reduces hydrogen absorption by the steel substrate **10**. Further, as the Al—Fe phase **22b** fixes hydrogen atoms, hydrogen absorption by the steel substrate **10** is reduced.

(17) Further, the present inventors discovered that hydrogen absorption by the steel substrate **10** can be further reduced if the oxide layer **23** contains predetermined amounts of one or more of Be, Mg, Ca, Sr, Ba, Sc and Zn (these seven elements will be hereinafter referred to as “Group B elements”). The Group B elements have the property of being oxidized more readily than Al and have the effect of reducing oxidation of Al to reduce hydrogen absorption by the steel substrate **10**.

(18) As the Fe—Al—Si phase **22a** of the intermediate layer **22** contains predetermined amounts of Group A elements and the oxide layer **23** contains predetermined amounts of Group B elements, their total effect provides a hot-stamped product **1** having improved delayed-fracture resistance.

(19) The present invention was made based on these discoveries. Now, embodiments of the present invention will be described in detail with reference to the drawings. The same or corresponding components in the drawings are labeled with the same reference numerals, and their description will not be repeated. The ratios between the components shown in the drawings do not necessarily represent the actual size ratios.

(20) [Hot-Stamped Product]

(21) Returning to FIG. **3**, the construction of a hot-stamped product **1** according to an embodiment of the present invention will be described. The hot-stamped product **1** is produced by forming steel material through hot stamping. The hot-stamped product **1** includes a steel substrate **10** and an Al film **20** formed on the steel substrate **10**.

(22) The Al film **20** includes an interface layer **21** located at the interface with the steel substrate **10**, an intermediate layer **22** formed on the interface layer **21**, and an oxide layer **23** formed on the intermediate layer **22**.

(23) [Interface Layer]

(24) The interface layer **21** is formed at the interface with the steel substrate **10**. The interface layer **21** is structured such that part of the bcc structure of the α Fe is substituted by Al and Si.

(25) The chemical composition of the interface layer **21** has a distribution that changes as it goes along the thickness direction. For example, the interface layer **21** has a chemical composition of, in average along the thickness direction: 60 to 98 mass % Fe; 1 to 40 mass % Al; and 1 to 20 mass % Si. The interface layer **21** may contain small amounts of elements other than Si, Fe, and Al. An

upper limit for the total content of the elements other than Si, Fe and Al contained in the interface layer **21** is preferably 3.0 mass %, and more preferably 1.5 mass %. In some cases, the border between the interface layer **21** and steel substrate **10** may not be clear; in such cases, the depth range from the interface with the intermediate layer **22** to 10 μm therefrom is treated as the interface layer **21**, and the average chemical composition in that range is determined.

(26) The chemical composition of the interface layer **21** may be measured by analyzing a cross section of the Al film **20** using an electron probe micro-analyzer (EPMA) or an energy dispersive spectrometer (EDS) of an SEM.

(27) The interface layer **21** is not limited to any particular thickness, and may be 5 to 15 μm thick, for example.

(28) [Intermediate Layer]

(29) The intermediate layer **22** is formed on the interface layer **21**. The intermediate layer **22** includes an Fe—Al—Si phase **22a** and an Al—Fe phase **22b**. As is the case with the interface layer **21**, the Fe—Al—Si phase **22a** is structured such that part of the bcc structure of the αFe is substituted by Al and Si. The Al—Fe phase **22b** has a structure of $\text{Fe}_{0.4}\text{Al}_{0.13}$ and $\text{Fe}_{0.2}\text{Al}_{0.5}$. Solute Si may be present in the Al—Fe phase **22b**, too.

(30) The Fe—Al—Si phase **22a** and Al—Fe phase **22b** may be identified by analyzing the chemical composition at a cross section of the Al film **20** using an EPMA or an EDS of an SEM. The Fe—Al—Si phase **22a** and Al—Fe phase **22b** may also be identified by analyzing the crystal structure through X-ray diffraction or electron diffraction.

(31) Although the interface layer **21** and Fe—Al—Si phase **22a** have similar structures, these two can be distinguished based on position. Specifically, the interface layer **21** is located at the interface with the steel substrate **10**, whereas the Fe—Al—Si phase **22a** is located in the middle of the intermediate layer **22** along the thickness direction.

(32) The Si content of the Fe—Al—Si phase **22a** is 1 to 20 mass %. If the Si content is lower than 1 mass %, the oxide layer **23** grows, which increases the amount of hydrogen that is absorbed in the steel substrate **10**. If the Si content is higher than 20 mass %, the intermediate layer **22** does not grow sufficiently, which, too, increases the amount of hydrogen that is absorbed in the steel substrate **10**. A lower limit for the Si content of the Fe—Al—Si phase **22a** is preferably 5 mass %, and more preferably 8 mass %. An upper limit for the Si content of the Fe—Al—Si phase **22a** is preferably 18 mass %, and more preferably 16 mass %.

(33) The Fe content of the Fe—Al—Si phase **22a** is not limited to any particular value, and may be 30 to 80 mass %, for example. The Al content of the Fe—Al—Si phase **22a** is not limited to any particular value, and may be 5 to 50 mass %, for example.

(34) In addition to Fe, Al and Si, the Fe—Al—Si phase **22a** contains one or more elements selected from the group consisting of Zr, Ce, Y, Ta, Ni, Cu, Nb, Cr, Co, V and Ti. The eleven elements Zr, Ce, Y, Ta, Ni, Cu, Nb, Cr, Co, V and Ti will be hereinafter referred to as Group A elements. Preferable Group A elements are Zr, Ce, Ni, Cr, Co, V and Ti, and particularly preferable ones are Ni and Cr.

(35) The total content of the Group A elements in the Fe—Al—Si phase **22a** is 0.10 to 5.0 mass %. The Group A elements reduce stabilization of the Fe—Al—Si phase **22a** to promote growth of the Al—Fe phase **22b**, thereby reducing hydrogen absorption by the steel substrate **10**. This effect is not sufficiently produced if the total content of the Group A elements is lower than 0.1 mass %. On the other hand, if the total content of the Group A elements is higher than 5.0 mass %, the Fe—Al—Si phase **22a** may become oversized. A lower limit for the total content of the Group A elements in the Fe—Al—Si phase **22a** is preferably 0.20 mass %, and more preferably 0.30 mass %. An upper limit for the content of the Group A elements in the Fe—Al—Si phase **22a** is preferably 4.0 mass %, and more preferably 3.0 mass %.

(36) The Fe—Al—Si phase **22a** may contain small amounts of elements other than Si, Fe, Al and the Group A elements. An upper limit for the total content of the elements other than Si, Fe, Al and

the Group A elements contained in the Fe—Al—Si phase **22a** is preferably 1.0 mass %, and more preferably 0.5 mass %.

(37) As is the case with the interface layer **21**, the chemical composition of the Fe—Al—Si phase **22a** may be measured by analyzing a cross section of the Al film **20** using an EPMA or an EDS of an SEM. Specifically, the content of a given element is measured at a plurality of locations, and the average is treated as the content of this particular element. The chemical composition of the Al—Fe phase **22b** discussed below can be measured by the same method.

(38) The Al—Fe phase **22b** is not limited to any particular chemical composition, and may have a composition of, for example, 15 to 70 mass % Fe, 30 to 85 mass % Al, and 0 to 20 mass % Si. As discussed above, the Al—Fe phase **22b** basically has a structure of Fe.sub.4Al.sub.13 and Fe.sub.2Al.sub.5, however, solute Si may be present. The Al—Fe phase **22b** may contain small amounts of elements other than Si, Fe, and Al. An upper limit for the total content of the elements other than Si, Fe, and Al contained in the Al—Fe phase **22b** is preferably 1.0 mass %, and more preferably 0.5 mass %.

(39) The area ratio of the Fe—Al—Si phase **22a** in the intermediate layer **22** is preferably 5 to 35%. There is a tendency that the lower the area ratio of the Fe—Al—Si phase **22a** in the intermediate layer **22**, the better hydrogen absorption by the steel substrate **10** will be reduced. An upper limit for the area ratio of the Fe—Al—Si phase **22a** in the intermediate layer **22** is more preferably 30%, and yet more preferably 25%.

(40) In addition to the Fe—Al—Si phase **22a** and Al—Fe phase **22b**, the intermediate layer **22** may include small amounts of other phases. An example of a phase other than the Fe—Al—Si phase **22a** and Al—Fe phase **22b** is τ phase. The area ratio of the phases other than the Fe—Al—Si phase **22a** and Al—Fe phase **22b** in the intermediate layer **22** is preferably not higher than 10.0%, and more preferably not higher than 5.0%.

(41) The thickness of the intermediate layer **22** is not smaller than 15 μm . The larger the thickness of the intermediate layer **22**, the better diffusion of Al in the Al film **20** toward its surface will be reduced such that oxidation of Al will be better reduced. Furthermore, hydrogen can be fixed in the intermediate layer **22** more easily. Thus, there is a tendency that the larger the thickness of the intermediate layer **22**, the better hydrogen absorption by the steel substrate **10** will be reduced. Further, there is a tendency that the larger the thickness of the intermediate layer **22**, the better coating adhesiveness and corrosion resistance. A lower limit for the thickness of the intermediate layer **22** is preferably 20 μm , and more preferably 25 μm . An upper limit for the thickness of the intermediate layer **22** is not limited to a particular value, and may be 50 μm , for example.

(42) [Oxide Layer]

(43) The oxide layer **23** is formed on the intermediate layer **22**. The oxide layer **23** is mainly composed of oxides of Al. The oxide layer **23** may contain elements other than Al and O. The oxide layer **23** may contain, for example, Si, Fe, and Group A elements discussed above.

(44) The oxide layer **23** further includes one or more elements selected from the group consisting of Be, Mg, Ca, Sr, Ba, Sc and Zn. The seven elements Be, Mg, Ca, Sr, Ba, Sc and Zn will be hereinafter referred to as Group B elements. Preferable Group B elements are Mg, Ca, Sr and Ba, and a particularly preferable one is Mg. The group B elements are preferably present as oxides.

(45) The total proportion of the Group B elements in the composition of the oxide layer **23** excluding oxygen (hereinafter simply referred to as “proportion of the Group B elements”) is to be 0.01 to 80.0 mass %. The Group B elements have the property of being oxidized more readily than Al, and has the effect of reducing oxidation of Al to reduce hydrogen absorption by the steel substrate **10**. This effect is not sufficiently produced if the proportion of the Group B elements is lower than 0.01 mass %. If the proportion of the Group B elements is higher than 80.0 mass %, the thickness of the oxide layer **23** may increase. A lower limit for the proportion of the Group B elements is preferably 1.0 mass %, more preferably 3.0 mass %, yet more preferably 10 mass %, and still more preferably 15 mass %. An upper limit for the proportion of the Group B elements is

preferably 60.0 mass %, more preferably 40.0 mass %, and yet more preferably 35 mass %.

(46) As is the case with the interface layer **21** and intermediate layer **22**, the chemical composition of the oxide layer **23** may be measured by analyzing a cross section of the Al film **20** using an EPMA or an EDS of an SEM. It is also possible with an EDS of a TEM. The chemical composition of the oxide layer **23** may also be measured through depth-wise composition analysis with a glow discharge spectrometer (GDS) or by Auger electron spectroscopic analysis (AES).

(47) The thickness of the oxide layer **23** is preferably 0.01 to 1.00 μm . There is a tendency that the smaller the thickness of the oxide layer **23**, the better hydrogen absorption by the steel substrate **10** will be reduced. An upper limit for the thickness of the oxide layer **23** is preferably 0.50 μm , and more preferably 0.30 μm .

(48) The interface between the oxide layer **23** and intermediate layer **22** can be determined by observing the distribution of oxygen concentration. In some of the present embodiments, a GDS is used and the position at which the detection intensity for oxygen is as low as $\frac{1}{6}$ of the maximum is determined to be the interface between the oxide layer **23** and intermediate layer **22**.

(49) Particularly preferable combinations of a Group A element and a Group B element are a combination of Ni and Mg and a combination of Cr and Mg. That is, it is particularly preferable that the Fe—Al—Si phase **22a** includes one or two elements selected from the group consisting of Ni and Cr, the total content of Ni and Cr in the Fe—Al—Si phase **22a** is 0.10 to 5.0 mass %, the oxide layer **23** includes Mg, and the proportion of Mg in the composition of the oxide layer **23** excluding oxygen is 0.01 to 80.0 mass %.

(50) [Steel Substrate]

(51) The steel substrate **10** is not limited to any particular steel as long as it is suitable for hot stamping. Examples of substrate steels that can be used in the hot-stamped product **1** include steels having a chemical composition of, in mass %: 0.1 to 0.6% C; 0.01 to 0.60% Si; 0.50 to 3.00% Mn; up to 0.05% P; up to 0.020% S; up to 0.10% Al; 0.01 to 0.10% Ti; 0.0001 to 0.0100% B; up to 0.010% N; 0 to 1.0% Cr; 0 to 1.0% Mo; 0 to 1.0% Cu; 0 to 1.0% Ni; and the balance Fe and impurities. Examples of forms of the steel substrate prior to hot stamping include steel sheet such as hot-rolled steel sheet and cold-rolled steel sheet. The chemical composition of the steel substrate **10** will be described below. In the following description, “%” for the content of an element means mass %.

(52) C: 0.1 to 0.6%

(53) Carbon (C) is included in steel to ensure that it has the intended mechanical strength. If C content is too low, a sufficient improvement in mechanical strength may not be provided. On the other hand, if C content is too high, elongation or reduction of area can easily decrease. An upper limit for C content is more preferably 0.4%.

(54) Si: 0.01 to 0.60%

(55) Silicon (Si) is an element that improves mechanical strength and, similar to C, is included in steel to ensure that it has the intended mechanical strength. If Si content is too low, a sufficient improvement in mechanical strength may not be provided. On the other hand, if Si content is too high, Si oxides formed at the surface layer of the steel substrate may effect a decrease in wettability during plating and thus leaving some portions unplated.

(56) Mn: 0.50 to 3.00%

(57) Manganese (Mn) is one of the elements that strengthen steel, i.e., strengthening elements, and is also one of the elements that enhance hardenability. Mn is also effective in reducing hot shortness caused by S, which is an impurity. These effects may not be sufficiently produced if Mn content is too low. On the other hand, if Mn content is too high, an excessive amount of residual austenite may result, leading to decreased strength.

(58) P: Up to 0.05%

(59) Phosphorus (P) is an impurity contained in a steel substrate. P contained in the steel substrate may segregate on the crystal grain boundaries of the steel substrate and thus decrease the toughness

of the steel substrate. It is preferable to keep P content as low as possible.

(60) S: Up to 0.020%

(61) Sulfur (S) is an impurity contained in a steel substrate. S contained in the steel substrate may form sulfides and decrease the toughness of the steel substrate. It is preferable to keep S content as low as possible.

(62) Al: Up to 0.10%

(63) Aluminum (Al) is typically used to deoxidize steel. However, if Al content is high, the Ac.sub.3 point of the steel substrate rises, which requires an increase in the heating temperature during hot stamping to ensure hardenability of the steel. In view of this, Al content is preferably not higher than 0.10%. Al content is more preferably not higher than 0.05%, and yet more preferably not higher than 0.01%.

(64) Ti: 0.01% to 0.10%

(65) Titanium (Ti) is one of the elements that increase strength. The effects of improving strength and oxidation resistance may not be sufficiently produced if Ti content is too low. On the other hand, if Ti content is too high, carbides and/or nitrides may be formed, softening the steel.

(66) B: 0.0001 to 0.0100%

(67) Boron (B) has the effect of acting during quenching to increase strength. The effect of improving strength may not be sufficiently produced if B content is too low. On the other hand, if B content is too high, inclusions may be formed, embrittling the steel substrate and decreasing its fatigue strength.

(68) N: Up to 0.010%

(69) Nitrogen (N) is an impurity contained in a steel substrate. N contained in the steel substrate may form nitrides and decrease the toughness of the steel substrate. Further, N contained in the steel substrate may combine with B and thus decrease the amount of solute B, decreasing B's effect of improving hardenability. It is preferable to keep N content as low as possible.

(70) The steel substrate **10** may also contain Cr, Mo, Cu and Ni.

(71) Cr: 0 to 1.0%

(72) Mo: 0 to 1.0%

(73) Cu: 0 to 1.0%

(74) Ni: 0 to 1.0%

(75) To improve the hardenability of the steel substrate, one or more elements selected from the group consisting of chromium (Cr), molybdenum (Mo), copper (Cu), and nickel (Ni) may be included. A preferred lower limit for the content of each of these elements is 0.01%. On the other hand, if excessive amounts are included, saturation will be reached in terms of effect, increasing costs.

(76) The balance of the chemical composition of the steel substrate **10** is Fe and impurities. Impurity as used herein means an element originating from ore or scrap used as raw material for steel or an element that has entered from the environment or the like during the manufacturing process. Examples of impurities other than the elements listed above are Zn, Co, Sn, Nb, V, As, Zr, Ca, and Mg.

(77) [Method of Manufacturing Hot-Stamped Product]

(78) Next, an exemplary method of manufacturing a hot-stamped product **1** will be described. The manufacturing method described below involves plating steel material, such as steel sheet, with Al to produce a plated steel member, and hot stamping the plated steel member to form an Al film **20** on the steel substrate **10**. The method described here is an illustrative example, and does not limit the method of manufacturing a hot-stamped product **1**.

(79) [Plating Step]

(80) Hot dipping is used to form a plating layer on the surface of steel material. The temperature of the plating bath is preferably 600 to 700° C. If the temperature of the plating bath is lower than 600° C., the plating bath has low viscosity, making uniform plating difficult. If the temperature of

the plating bath is higher than 700° C., volatilization occurs to change the composition in a short period of time, making the process management difficult.

(81) The addition of Group A elements is performed by adding them to the plating bath. The content of the Group A elements in the plating bath is preferably 0.05 to 5.0 mass %. There is a tendency that the higher the content of the Group A elements in the plating bath, the higher the content of the Group A elements in the Fe—Al—Si phase **22a**. A lower limit for the content of the Group A elements in the plating bath is more preferably 0.2 mass %, and yet more preferably 0.5 mass %. An upper limit for the content of the Group A elements in the plating bath is more preferably 3.0 mass %, and yet more preferably 2.0 mass %.

(82) The addition of Group B elements is also performed by adding them to the plating bath. If Group B elements are to be added, the content of the Group B elements in the plating bath is preferably 0.01 to 1.0 mass %. There is a tendency that the higher the content of the Group B elements in the plating bath, the higher the content of the Group B elements in the oxide layer **23**. A lower limit for the content of the Group B elements in the plating bath is more preferably 0.05 mass %, and yet more preferably 0.08 mass %. An upper limit for the content of the Group B elements in the plating bath is more preferably 0.8 mass %, and yet more preferably 0.6 mass %.

(83) The Si content of the plating bath is 1.0 to 20.0 mass %, for example. The balance of the plating bath is mainly Al. The plating bath may contain small amounts of elements other than Al, Si and the Group A and B elements. The total contained of the elements other than Al, Si and the Group A and B elements contained in the plating bath is preferably not higher than 5.0 mass %, more preferably not higher than 3.0 mass %, yet more preferably not higher than 2.0 mass %, and still more preferably not higher than 1.0 mass %.

(84) The steel material is held in a hydrogen reduction atmosphere at 700 to 800° C. before being immersed in the plating bath. It is preferable to perform the entire process in a non-oxidizing atmosphere. The plating is preferably such that the resulting thickness of the plating layer is 20 to 30 μm. The thickness of the plating layer is adjusted such that the thickness of the intermediate layer **22** after the step of hot stamping is not smaller than 15 μm. The thickness of the plating layer can be adjusted by regulating the temperature and viscosity of the plating bath, immersion time, and/or gas blowing.

(85) [Hot-Stamping Step]

(86) The plated steel member is shaped to the required size before being hot stamped. The heating may be performed in a hot-temperature furnace or through electrical heating. The rate of temperature increase may be 1 to 50° C./s, for example. It is preferable that the retention temperature is 850 to 950° C. and it is preferable that the holding time is not shorter than 2 minutes. The rate of temperature decrease (i.e., cooling) may be 30 to 1000° C./s, for example.

(87) If the holding time is too long, the crystal structure of the intermediate layer may change. An upper limit for the holding time is preferably 30 minutes, and more preferably 10 minutes.

(88) The above-described process results in the hot-stamped product **1**. Although a method of hot stamping a plated steel member has been described, a hot-stamped product **1** may be produced by using vapor deposition or spraying to cause Al and/or other elements to adhere to the surface of the steel substrate **10** to form an Al coating layer, and then hot stamping this steel substrate **10** having the Al coating layer.

(89) These embodiments provide a hot-stamped product with improved delayed-fracture resistance.

EXAMPLES

(90) Now, the present invention will be described more specifically with reference to examples. The present invention is not limited to these examples.

(91) Steel sheets having the chemical compositions shown in Table 1 were prepared, and each of the steel sheets was subjected to hot dipping to form an Al—Si plating layer on each side thereof.

(92) TABLE-US-00001 TABLE 1 Steel Chemical composition (in mass %, balance Fe and impurities)

No.	C	Si	Mn	P	S	Al	Ti	B	N	S1	0.1	0.21	1.21	0.02	0.005	0.05	0.02	0.0030	0.005	S2	0.4
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0.01 1.01 0.04 0.010 0.03 0.04 0.0022 0.004

(93) The plating baths used had the chemical compositions shown in Table 2. The temperature of the plating baths, during hot dipping was 700° C. After each steel sheet was immersed in the associated plating bath, the amount of adhesion was adjusted by gas wiping to 70 g/m² for each side.

(94) TABLE-US-00002 TABLE 2 Chemical composition of plating bath Steel (in mass %, balance Al and impurities) Table No. Si Group A elements Group B elements A01 S1 10.0 Co: 0.20 Ca: 0.10 A02 S2 10.0 Zr: 1.00 Mg: 0.1 A03 S1 10.0 Ni: 2.00 Ba: 0.1 A04 S2 10.0 Co: 1.00 Sr: 0.1 A05 S1 10.0 V: 0.05 Mg: 0.05 A06 S2 10.0 V: 0.50 — A07 S1 10.0 Nb: 0.50 — A08 S2 10.0 Nb: 2.00 — A09 S1 10.0 Y: 0.10 — A10 S2 10.0 Cr: 0.50 Mg: 0.10 A11 S1 10.0 Ni: 2.00 — A12 S2 10.0 Cr: 1.00 Ca: 0.05 A13 S1 10.0 Ni: 0.50 Mg: 0.10 A14 S1 10.0 Cu: 1.00 — A15 S1 10.0 Ce: 0.20 Mg: 0.05 A16 S2 10.0 Ta: 0.50 — A17 S2 10.0 Ti: 0.50 Ca: 0.05 a1 S1 10.0 — — a2 S1 10.0 Zr: 0.005 — a3 S1 10.0 — Mg: 0.60 a4 S1 10.0 Y: 0.05 Ca: 0.50 a5 S2 10.0 — Ca: 0.005 a6 S2 10.0 Co: 0.005 — a7 S2 10.0 Ni: 7.00 —

(95) The plated steel sheet was heated in an electric resistance furnace at a furnace temperature of 900° C. under a synthetic-air flow with a dew point of 20° C. in such a manner that the soaking time was five minutes. Thereafter, the sheet was formed in a die and, at the same time, cooled in the die to provide a hot-stamped product.

(96) The resulting hot-stamped products were examined for the Si content of the Fe—Al—Si phase, the content of the Group A elements in the Fe—Al—Si phase, the area ratio of the Fe—Al—Si phase in the intermediate layer, the thickness of the intermediate layer, the proportion of the Group B elements in the composition of the oxide layer excluding oxygen, and the thickness of the oxide layer. Further, the following properties were evaluated: hydrogen embrittlement resistance (i.e., delayed-fracture resistance), coating adhesiveness, post-coating corrosion resistance, and pitting resistance.

(97) [Hydrogen Embrittlement Resistance]

(98) The resulting hot-stamped products were analyzed for hydrogen. The hydrogen analysis was performed by temperature-programmed desorption, where diffusible hydrogen was defined as hydrogen emitted until 250° C. was reached and a score was given based on its amount in accordance with the following criteria: 2: amount of diffusible hydrogen smaller than 0.1 mass ppm 1: amount of diffusible hydrogen not smaller than 0.1 mass ppm and smaller than 0.2 mass ppm 0: amount of diffusible hydrogen not smaller than 0.2 mass ppm

[Coating Adhesiveness]

(99) Coating adhesiveness was evaluated in accordance with the method disclosed in Japanese Patent No. 4373778. That is, each specimen was immersed in deionized water at 60° C. for 240 hours before a grid was cut therein using a cutter, with a distance of 1 mm between adjacent cuts and having 100 squares, and the number of squares in the peeled region was visually measured to calculate the area ratio, based on which a score was given in accordance with the following criteria: 3: area of peeling of 0% or larger but smaller than 10% 2: area of peeling not smaller than 10% and smaller than 70% 1: area of peeling not smaller than 70% up to 100%

[Post-Coating Corrosion Resistance]

(100) Corrosion resistance after coating was evaluated by the method specified by JASO M609 established by the Society of Automotive Engineers of Japan, Inc. The coating was scratched using a cutter; after 180 cycles of corrosion testing, the width with coating expansion as measured from a scratch (the maximum on a single side) was measured; and a score was given in accordance with the following criteria: 3: width with expansion of 0 mm or larger but smaller than 1.5 mm 2: width with expansion not smaller than 1.5 mm and smaller than 3.0 mm 1: width with expansion not smaller than 3.0 mm

[Pitting Resistance]

(101) Each specimen was immersed in the PREPALENE-X surface conditioner from Nihon

Parkizing Co., Ltd. for one minute at room temperature, before being immersed in the PALBOND SX35 coating substrate formation agent from the same company for two minutes at 35° C. Thereafter, the specimen was subjected to combined cyclic corrosion testing by the method specified by JIS H 8502. The specimen was coated with POWERFLOAT 1200 from Nippon Paint Co., Ltd. to a thickness of 15 μm, and cut using a cutter as specified in JIS H 8502. A score was given based on the amount of reduction in the sheet thickness of the steel sheet after 60 cycles, measured at cut portions, in accordance with the following criteria: 5: amount of reduction in sheet thickness smaller than 0.1 mm 4: amount of reduction in sheet thickness not smaller than 0.1 mm and smaller than 0.2 mm 3: amount of reduction in sheet thickness not smaller than 0.2 mm and smaller than 0.3 mm 2: amount of reduction in sheet thickness not smaller than 0.3 mm and smaller than 0.4 mm 1: amount of reduction in sheet thickness not smaller than 0.4 mm

(102) The results are shown in Table 3.

(103) TABLE-US-00003 TABLE 3 Intermediate Layer Content of Group A Oxide Layer Si content elements Area ratio Proportion of of Fe—Al—Si in Fe—Al—Si of Fe—Al—Si Group B phase phase phase Thickness elements Thickness Evaluation Lable (mass %) (mass %) (%) (μm) (mass %) (μm) E1* E2* E3* E4* Total A01 14 0.1 20 20 20.0 0.15 2 3 2 4 11 A02 15 0.5 15 21 25.0 0.18 2 3 3 5 13 A03 15 1.5 10 25 20.0 0.15 2 3 3 5 13 A04 15 0.2 15 25 15.0 0.15 2 3 3 5 13 A05 12 0.1 25 20 20.0 0.15 2 3 3 3 11 A06 15 0.5 20 23 — 0.15 1 3 3 3 10 A07 12 0.2 25 20 — 0.10 1 3 3 4 11 A08 13 0.5 15 23 — 0.20 1 3 3 3 10 A09 10 0.1 30 20 — 0.15 1 3 3 5 12 A10 15 0.5 15 25 25.0 0.18 2 3 3 5 13 A11 15 2.0 10 30 — 0.20 1 3 3 5 12 A12 15 1.0 15 25 15.0 0.22 2 3 3 5 13 A13 16 1.0 15 30 20.0 0.20 2 3 3 5 13 A14 14 1.0 15 23 — 0.15 1 3 3 5 12 A15 11 0.1 20 20 20.0 0.15 2 3 3 4 11 A16 13 0.2 15 25 — 0.30 1 3 1 5 12 A17 15 0.3 10 23 20.0 0.25 2 3 1 5 13 a1 10 — 40 20 — 0.32 0 2 1 2 5 a2 12 <0.01 38 20 — 0.35 0 3 1 2 6 a3 10 — 40 20 82.0 0.35 0 2 1 2 5 a4 12 0.05 38 25 65.0 1.00 0 3 1 2 6 a5 12 — 38 20 <0.01 0.30 0 1 1 2 4 a6 10 <0.01 38 20 — 0.30 0 1 1 2 4 a7 7 5.5 40 15 — 0.30 0 1 1 2 4 E1*: Hydrogen embrittlement resistance E2*: Coating adhesiveness E3*: Post-coating corrosion resistance E4*: Pitting resistance

(104) As shown in Table 3, for each of the hot-stamped products labeled A01 to A17, the Si content of the Fe—Al—Si phase was in the range of 1 to 20 mass %, and the content of Group A elements was in the range of 0.10 to 5.0 mass %. For each of these hot-stamped products, the amount of diffusible hydrogen in hydrogen-embrittlement resistance testing was smaller than 0.2 mass ppm. These hot-stamped products had good coating adhesiveness, post-coating corrosion resistance and pitting resistance. For the hot-stamped products labeled A01 to A05, A10, A12, A13, A15 and A17 in which the oxide layer contained Group B elements, the amount of diffusible hydrogen in hydrogen embrittlement resistance testing was smaller than 0.1 mass ppm, which means particularly good hydrogen embrittlement resistance (i.e., delayed-fracture resistance).

(105) For each of the hot-stamped products labeled a1 to a7, the amount of diffusible hydrogen in hydrogen embrittlement resistance testing was not smaller than 0.2 mass ppm, which means that they had a lower hydrogen embrittlement resistance (delayed-fracture resistance) than the hot-stamped products labeled A01 to A20. This is presumably because, for the hot-stamped products labeled a1 to a7, the content of the Group A elements in the Fe—Al—Si phase was either lower than 0.1 mass % or higher than 5.0 mass %.

(106) FIG. 4 is a photograph showing a cross section of the Al film of the product labeled a1. FIG. 5 is a photograph of a cross section of the Al film of the product labeled A13. As shown in FIGS. 4 and 5, for the product labeled A13, the area ratio of the Fe—Al—Si phase was lower than that for the product labeled a1. Further, the Fe—Al—Si phase was discontinuous, whereas that of the product labeled a1 had the shape of a layer (i.e., band).

(107) Although embodiments of the present invention have been described, the above-described embodiments are merely illustrative examples useful for carrying out the present invention. Thus, the present invention is not limited to the above-described embodiments, and the above-described embodiments, when carried out, may be modified as appropriate without departing from the spirit

of the invention.

REFERENCE SIGNS LIST

(108) **1, 9**: hot-stamped product **1A**: plated steel member **10**: steel substrate **20**: Al film **21**: interface layer **22**: intermediate layer **22a**: Fe—Al—Si phase **22b**: Al—Fe phase **23**: oxide layer **30**: Al—Si plating layer **35**: Al—Si—Fe alloy layer

Claims

1. A hot-stamped product comprising: a steel substrate; and an Al film formed on the steel substrate, the Al film including: an interface layer located at an interface with the steel substrate and having a structure with part of α Fe substituted by Al and Si; an intermediate layer formed on the interface layer; and an oxide layer formed on the intermediate layer, a thickness of the intermediate layer being not smaller than 15 μm , the intermediate layer including an Fe—Al—Si phase having a structure with part of α Fe substituted by Al and Si, the Fe—Al—Si phase including one or more elements selected from the group consisting of Zr, Ce, Y, Ta, Ni, Cu, Nb, Co, and V, an Si content of the Fe—Al—Si phase being 1 to 20 mass %, a total content of Zr, Ce, Y, Ta, Ni, Cu, Nb, Co, and V being 0.10 to 5.0 mass %, the oxide layer including one or more elements selected from the group consisting of Be, Mg, Ca, Sr, Ba, Sc and Zn, a total proportion of Be, Mg, Ca, Sr, Ba, Sc and Zn in a composition of the oxide layer excluding oxygen being 0.01 to 80.0 mass %.
 2. The hot-stamped product according to claim 1, wherein an area ratio of the Fe—Al—Si phase in the intermediate layer is 5 to 35%.
 3. The hot-stamped product according to claim 1, wherein a thickness of the oxide layer is 0.01 to 1.0 μm .
 4. The hot-stamped product according to claim 1, wherein: a content of Ni in the Fe—Al—Si phase is 0.10 to 5.0 mass %; and a proportion of Mg in a composition of the oxide layer excluding oxygen is 0.01 to 80.0 mass %.
 5. The hot-stamped product according to claim 3, wherein: a content of Ni in the Fe—Al—Si phase is 0.10 to 5.0 mass %; and a proportion of Mg in a composition of the oxide layer excluding oxygen is 0.01 to 80.0 mass %.
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